

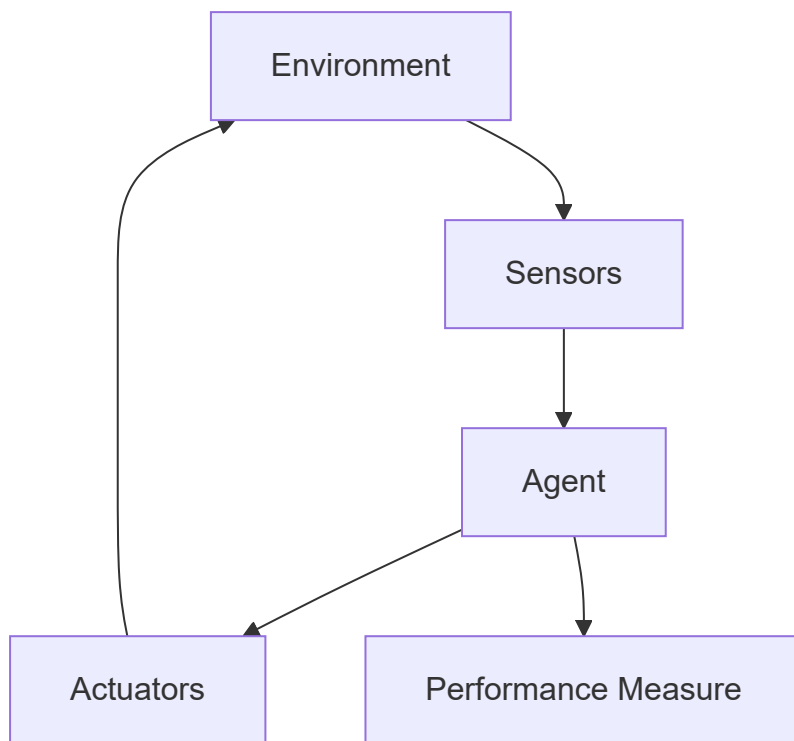
AI Module 1

What is a rational agent? Explain with diagram. (5 marks)

A rational agent is an AI system that acts to achieve the best possible outcome or the best expected outcome given its knowledge and capabilities. It perceives its environment through sensors and acts upon it through actuators.

Key properties of a rational agent:

- Performs actions that maximize performance measure
- Uses percept history and built-in knowledge
- Acts optimally based on available information



Explain PEAS and also write down the PEAS representations for Medical diagnosis system. (10 marks)

PEAS is a framework for describing task environments in AI. It stands for Performance measure, Environment, Actuators, and Sensors.

PEAS components:

- **Performance measure:** Criteria that define success
- **Environment:** Context where agent operates

- **Actuators:** Devices that affect the environment
- **Sensors:** Devices that perceive the environment

PEAS representation for Medical diagnosis system:

Component	Description
Performance measure	Accurate diagnosis, minimal false positives/negatives, fast response time, patient safety
Environment	Patient, symptoms, medical history, test results, hospital database, medical literature
Actuators	Display diagnosis, prescribe treatment, recommend tests, alert doctor, update patient record
Sensors	Keyboard input, symptom questionnaire, vital signs monitor, lab result interface, imaging data

State PEAS of automated taxi driver. (5 marks)

PEAS for automated taxi driver:

Component	Description
Performance measure	Safe driving, reaching destination, passenger comfort, fuel efficiency, obey traffic rules, minimize travel time
Environment	Roads, traffic signals, pedestrians, other vehicles, weather conditions, road signs, passengers
Actuators	Steering wheel, accelerator, brakes, indicators, horn, headlights, GPS navigation
Sensors	Cameras, radar, LIDAR, GPS, wheel speed sensors, accelerometer, microphones

Explain learning agents with example. (5 marks)

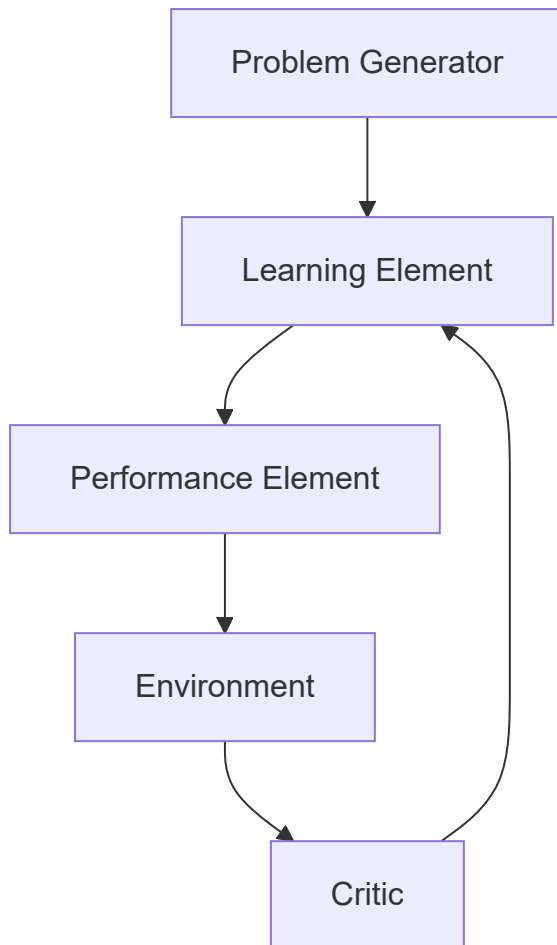
A learning agent improves its performance over time by acquiring knowledge from experience. It consists of four components: learning element, performance element, critic, and problem generator.

Example - Spam filter agent:

- **Learning element:** Updates classification rules based on user feedback

- **Performance element:** Decides whether incoming email is spam
- **Critic:** Compares classification to user marking (spam/not spam)
- **Problem generator:** Suggests borderline cases for user review

The agent initially classifies poorly but improves accuracy as it learns from labeled examples.



What are the different types of environments? Give examples. Explain the vacuum world problem with its environment. (10 marks)

Types of environments in AI:

Type	Description	Example
Fully observable	Agent sees complete environment state	Chess game
Partially observable	Agent sees limited information	Poker game
Deterministic	Next state fully determined by current action	Traffic light control
Stochastic	Random elements affect outcomes	Weather prediction

Type	Description	Example
Episodic	Each episode independent of previous	Image classification
Sequential	Current action affects future decisions	Robot navigation
Static	Environment doesn't change during reasoning	Sudoku puzzle
Dynamic	Environment changes while agent thinks	Stock trading
Discrete	Limited number of distinct states	Checkers game
Continuous	Infinite possible states	Medical diagnosis
Single-agent	Only one agent operates	Personal assistant
Multi-agent	Multiple agents interact	Autonomous vehicles

Vacuum World Problem:

The vacuum world has a rectangular grid with dirt in some squares. The vacuum cleaner agent can move left, move right, suck dirt, or do nothing.

Environment characteristics for vacuum world:

- Partially observable (agent sees only current square)
- Deterministic (actions have predictable results)
- Sequential (future dirt depends on current cleaning)
- Static (dirt doesn't reappear spontaneously in simple version)
- Discrete (grid positions and dirt states are countable)
- Single-agent (one vacuum cleaner)

Goal: Clean all dirt in minimum time/actions.

Differentiate between Simple Reflex Agent and Model-Based Reflex Agent. (5 marks)

Aspect	Simple Reflex Agent	Model-Based Reflex Agent
Internal state	No internal memory	Maintains internal state/model
Decision basis	Only current percept	Current percept + internal state
World knowledge	No knowledge of unobserved aspects	Tracks aspects not currently visible
Environment requirement	Fully observable required	Works with partial observability

Aspect	Simple Reflex Agent	Model-Based Reflex Agent
Complexity	Simple, fast	More complex, uses more memory
Example	Thermostat	Self-driving car tracking unseen objects

Discuss Problem Formulation in AI with an example. (5 marks)

Problem formulation is the process of defining a task in a structured way that an AI agent can solve. It converts a real-world situation into a formal search problem.

Components of problem formulation:

- **Initial state:** Starting point of the problem
- **Actions:** Possible moves from any state
- **Transition model:** Result of applying an action
- **Goal test:** Determines if target state is reached
- **Path cost:** Measure of solution quality

Example - Route finding from A to B:

- **Initial state:** Current location (A)
- **Actions:** Drive on connected roads
- **Transition model:** New location after driving
- **Goal test:** Reached destination (B)
- **Path cost:** Distance or travel time

Formulate the problem for 8 queens. (5 marks)

8 Queens Problem Formulation:

- **Initial state:** Empty 8×8 chessboard (no queens placed)
- **Actions:** Place a queen in any safe square of the next empty row (or column, depending on formulation)
- **Transition model:** New board state with one additional queen placed at the chosen position
- **Goal test:** All 8 queens are placed on the board such that no two queens attack each other (different rows, columns, and diagonals)

- **Path cost:** Number of queens placed (typically equal cost of 1 per placement) or constant cost focusing on finding any valid configuration

Alternative formulation (incremental):

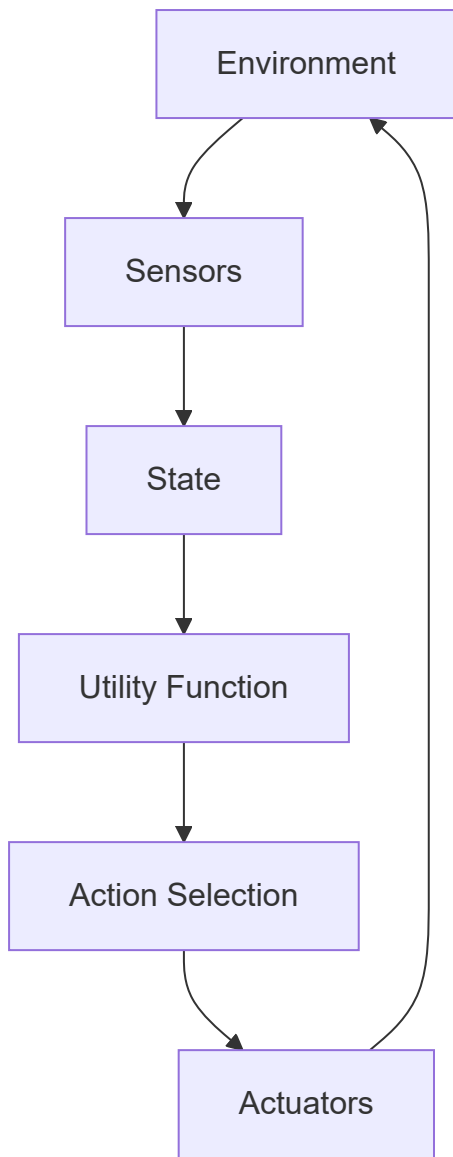
- **States:** Any arrangement of 0 to 8 queens with no attacks
 - **Initial state:** Empty board
 - **Goal test:** 8 queens with no mutual attacks
-

Explain utility based agent architecture with diagram. (5 marks)

A utility-based agent selects actions that maximize expected utility, a numerical measure of happiness or success. It handles trade-offs between conflicting goals.

Architecture components:

- Sensors perceive environment state
- Utility function calculates desirability of each possible outcome
- Decision component selects action maximizing expected utility



Example: A rescue robot choosing between saving two people assigns higher utility to saving a child than an adult, or considers both equally based on ethical rules.

Explain PEAS descriptors. Provide PEAS descriptors for Part Picking robot. (5 marks)

PEAS descriptors define the four components of an AI agent's task environment.

PEAS for Part Picking robot:

Component	Description
Performance measure	Pick correct parts, minimize pick time, avoid damage to parts, successful placement at destination
Environment	Conveyor belt, bins with parts, storage containers, obstacles, lighting conditions, other robots

Component	Description
Actuators	Robotic arm, gripper, suction cup, conveyor belt controller, camera positioning motor
Sensors	Vision camera, proximity sensors, force sensors, part identification sensor, position encoders

Explain goal based agents with example. (5 marks)

A goal-based agent selects actions to achieve a specific desired goal. It considers future consequences and plans sequences of actions.

Working:

- Maintains internal representation of world state
- Has defined goal (description of desired situation)
- Searches for action sequence from current state to goal
- Executes actions and updates state based on percepts

Example - Navigation robot:

- **Goal:** Reach room 105 from current location
- **Actions:** Turn left, turn right, move forward
- **Process:** Robot considers multiple paths, chooses efficient route, adjusts when encountering obstacles
- **Difference from reflex agent:** Plans ahead rather than reacting only to current sensor input

Explain state space based problem formulation. Formulate 8 Puzzle and N Queens problem. (10 marks)

State Space Problem Formulation:

State space formulation represents a problem as a graph where nodes are states and edges are actions. The solution is a path from initial state to goal state.

Components of state space:

- Set of all possible states
- Initial state (starting node)
- Actions applicable in each state (edges)
- Transition function (results of actions)

- Goal state or goal test
- Path cost function

8 Puzzle Formulation:

- **State representation:** 3×3 grid containing numbers 1-8 and one blank (0)
- **Initial state:** Any shuffled configuration
- **Actions:** Move blank up, down, left, right (if within bounds)
- **Transition model:** Blank swaps with adjacent tile
- **Goal state:** Ordered grid 1,2,3,[4,5,6],[7,8,0]
- **Path cost:** 1 per move

N Queens Problem Formulation:

- **State representation:** Array of N positions (column index for each row)
- **Initial state:** Empty board (no queens placed)
- **Actions:** Place queen in next row at any column not attacking existing queens
- **Transition model:** Add queen at (row, column) with attack checking
- **Goal test:** N queens placed with no mutual attacks
- **Path cost:** Constant per placement (usually 1)

Comparison table:

Aspect	8 Puzzle	N Queens
State space size	$9!/2 = 181,440$	Approximately $N!$
Branching factor	2-4 moves	$N - \text{row} + 1$
Goal condition	Exact configuration match	Constraint satisfaction
Solution type	Optimal path	Any valid configuration

What are the different types of environments? Give examples. Explain the vacuum world problem with its environment. (10 marks)

Types of environments in AI:

Environment Type	Description	Example
Fully vs Partially observable	Complete vs limited state information	Chess vs Poker

Environment Type	Description	Example
Deterministic vs Stochastic	Predictable vs random outcomes	Traffic light vs Weather
Episodic vs Sequential	Independent vs dependent decisions	Image recognition vs Robot navigation
Static vs Dynamic	Unchanging vs changing during deliberation	Sudoku vs Real-time game
Discrete vs Continuous	Countable vs infinite states	Checkers vs Self-driving car
Single vs Multi-agent	One agent vs multiple interacting agents	Personal assistant vs Autonomous vehicles

Vacuum World Problem:

The vacuum world consists of a 2D grid (commonly 2 squares or 4 squares). Each square may contain dirt or be clean. The vacuum cleaner perceives its current square's dirt status.

Environment characteristics:

- Partially observable (cannot see other squares)
- Deterministic (sucking cleans, moving changes location predictably)
- Sequential (cleaning order affects future dirty squares)
- Static (dirt doesn't appear spontaneously in standard version)
- Discrete (limited positions and dirt states)
- Single-agent (one vacuum cleaner)

Agent actions:

- Move left
- Move right
- Suck (removes dirt from current square)
- No operation (idle)

Performance metrics:

- Percentage of clean squares after N actions
- Total time to clean all squares
- Energy consumed

